

TQF ACADEMIC SERIES

TQF Quant Pathway Study Checklist

A compact review list based on the official TQF Quant Pathway page

Use this booklet to track reading coverage and topic familiarity.

Checklist Overview

Each section below groups the official Quant Pathway topics into a compact review format. Mark items as studied, revised, or ready for deeper work.

Foundational

Finance

- ☐ Time value of money (discounting, compounding, annuities)
- ☐ Financial instruments (equities, bonds, derivatives basics)
- ☐ Portfolio theory basics (CAPM, mean-variance, APT)
- ☐ Corporate finance basics (capital structure, leverage, WACC)
- ☐ Financial statements (balance sheet, P&L, cash flow)

Mathematics

- ☐ Calculus (differentiation, integration, multivariable calculus)
- ☐ Linear algebra (matrices, eigenvalues, SVD, optimization foundations)
- ☐ Probability and statistics (distributions, estimators, hypothesis testing, regression)
- ☐ Differential equations (ODEs, PDEs)

Programming

- ☐ Data Handling and Processing
- ☐ Statistical and Mathematical Libraries
- ☐ Object-Oriented Programming
- ☐ Numerical Methods
- ☐ Languages: Python, C++, R, VBA

Core

Advanced Mathematical Methods

- [] Stochastic calculus: Stochastic processes, Markov processes, martingales, Ito's Lemma, change of numeraire
- [] Statistical methods: Bayesian inference, copulas, and extreme value theory
- [] Numerical methods: Finite difference methods, Monte Carlo simulations, and variance reduction

Asset Behavior & Modeling

- [] Asset pricing (no-arbitrage, risk-neutral valuation, CAPM extensions)
- [] Option pricing theory (Binomial models, Black-Scholes, Greeks, implied volatility)
- [] Volatility modeling (ARCH/GARCH, stochastic volatility, SABR, volatility skews/smiles, realized volatility)
- [] Market microstructure basics (order book dynamics, liquidity, transaction costs, market impact models)

Fixed Income and Credit

- [] Term structure modeling (Vasicek, CIR, Hull-White, HJM, LMM, Nelson-Siegel)
- [] Yield curve construction and analysis (bootstrapping, interpolation techniques)
- [] Duration and convexity (key rate duration, partial durations)
- [] Credit risk models (structural vs reduced-form, Merton model, KMV)
- [] IFRS 9 / Basel frameworks (PD, LGD, EAD, ECL, stress testing)
- [] Securitization and structured products (ABS, MBS, CDOs, CLOs)

Risk Management

- [] Value-at-Risk (parametric, historical, Monte Carlo, tail VaR)
- [] Expected Shortfall (coherent risk measures)
- [] Hedging strategies (delta, gamma, vega, cross-asset hedging)
- [] Liquidity risk, counterparty risk, model risk (model validation, backtesting)
- [] Stress testing & scenario analysis (extreme events, reverse stress testing)

Machine Learning & Data Science

- [] Regression, classification, clustering
- [] Dimensionality reduction (PCA, SVD)
- [] Time series forecasting
- [] Feature engineering for financial data
- [] NLP (sentiment analysis, regulatory texts)
- [] Reinforcement learning in trading strategies

Specialized

Portfolio Management

- [] Factor models (Fama-French, risk premia, multi-factor models)
- [] Portfolio optimization (mean-variance, robust optimization, Black-Litterman model)
- [] Efficient frontier and Sharpe ratio (risk-adjusted performance metrics)
- [] Risk parity, smart beta, asset allocation (dynamic asset allocation, ESG investing)
- [] Performance attribution (Brinson model, multi-period attribution)
- [] Portfolio stress testing (scenario-based, factor-based, dynamic hedging strategies)

Trading & Execution

- [] Algorithmic trading (execution algos, TCA, market impact models)
- [] Market-making models (inventory management, bid-ask spread optimization)
- [] High-frequency trading (latency, order book modeling, co-location)
- [] Slippage and market impact (VWAP, TWAP, implementation shortfall)
- [] Statistical arbitrage and pairs trading

Derivatives and Advanced Products

- [] XVA (CVA, DVA, FVA, KVA, MVA, regulatory capital adjustments)
- [] Exotic options pricing (barrier, Asian, quanto, lookback, digital options)
- [] Real options & commodities (energy markets, real asset valuation)
- [] Local and stochastic volatility models (Dupire, Heston, SABR extensions)
- [] Interest rate and currency derivatives (swaps, caps, floors, cross-currency swaps)
- [] Hybrid derivatives (e.g., equity-credit hybrids)
- [] Weather and catastrophe derivatives

Credit Risk Management

- [] Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD)
- [] Credit portfolio modeling and forecasting
- [] Credit scoring/rating and early warning system (Machine learning applications)
- [] Counterparty risk and credit valuation adjustment (CVA)
- [] Stress testing for credit portfolios under regulatory frameworks

Regulation & Implementation

- [] Basel III calculations, scenario analysis, capital optimization
- [] IFRS-related, e.g. expected credit loss, hedge accounting, fair value
- [] Collateral management (CSA, ISDA, margining, collateral optimization)

[] Treasury and ALM models (liquidity coverage ratio, net stable funding ratio)

[] Regulatory capital optimization (RWA, leverage ratio, stress testing frameworks)